

## ABSTRACT

### Real-Time Gym Form Correction and Feedback System

AlignAI is an AI-driven fitness assistant designed to provide real-time posture correction and performance feedback for gym users. Using live telecast from the user's device camera, the app employs advanced computer vision techniques to detect body keypoints, calculate joint angles, and analyze overall movement patterns. When incorrect posture or form deviations are detected, the system instantly alerts the user through a beep sound and visual indicators, highlighting the exact body region that requires adjustment.

To ensure medically accurate guidance, the app is supported by a curated database of exercise images and demonstration videos certified by orthopedic specialists and physiotherapists. This database serves as the benchmark for correct posture, enabling the system to not only detect errors but also guide the user toward safe and effective exercise execution.

In addition, the application provides a personalized user dashboard where individuals can view their live session analytics, exercise history, posture correction reports, and improvement trends over time. The integration of audio-visual feedback ensures a seamless experience, allowing users to simultaneously listen to their own workout music while receiving corrective guidance.

By combining the expertise of medical professionals with cutting-edge AI, this application aims to reduce injury risks, enhance workout efficiency, and make posture correction accessible to gym-goers.